Automated Dynamic Analysis Signature Generation

Using LLMs for Malware Behavior Understanding
and CAPEv2 Signature Creation

Ikram Benfellah
Fadel Fatima Zahra

College of Computing, UM6P

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Outline

The Analyst Bottleneck

Malware detection is critical, but signature generation remains a time-consuming bottleneck.

Status quo

- 4–6 hours per sample
- Manual pattern extraction
- Inconsistent quality
- Difficult to scale

Desired state

- Minutes for initial drafts
- Consistent, standardized logic
- Human review preserved
- Ready to scale

Research Question

Can LLMs help analysts generate CAPEv2 signatures faster and with sufficient precision for deployment?

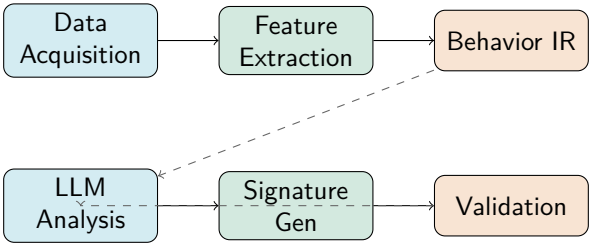
Project Scope and Constraints

- **Data:** 100 labeled malware samples from 10 families
- **Input:** CAPEv2 sandbox reports (JSON with behavioral telemetry)
- **Output:** Python signatures and MITRE ATT&CK mappings
- **Requirement:** Local inference (no cloud APIs, cost-conscious)

Success Criteria

Precision $\geq$ 80%, sufficient generalization, analyst workload reduction of 60%+.

System Architecture: Six-Stage Pipeline



Core Design Principle

Separate **data normalization** from **LLM reasoning**. Raw CAPEv2 JSON is noisy and inconsistent. A structured Behavior Intermediate Representation (IR) reduces hallucination and improves reproducibility.

Stage 1-3: Data Normalization

Why we built a Behavior IR: Raw CAPEv2 reports are Rich but messy.

Extracted signals

- Process creation & injection
- Registry modifications
- File I/O and dropped files
- Network & C2 beacons
- Executed commands

Benefits

- Reduces prompt noise
- Handles schema inconsistencies
- Enables family-level patterns
- $\sim$ 40% improvement in LLM precision

This abstraction layer is where robustness is built, not in the LLM.

Stage 4: LLM-Assisted Signature Synthesis

Tool: Llama 2 (7B), deployed locally on CPU

Why Llama 2?

- Open-source, no API costs
- Good for code generation
- Small enough for local inference
- Apache 2.0 license

Our implementation

- Feed normalized Behavior IR
- Generate Python signature drafts
- Family-level pattern aggregation
- Caching to reduce inference calls

LLM output is a *starting point*, never deployment-ready. Three validation gates are mandatory.

Stage 5-6: Validation Gates

We don't trust LLM output. Every signature goes through three sequential checks.

ate 1: **Syntax validation** — Python compiles, CAPEv2 API calls are valid

ate 2: **Logic validation** — Behavior checks match the IR; no contradictions

ate 3: **Manual review** — Analyst assesses family knowledge, refines MITRE mappings

Validation Impact

$\approx 15\%$ of generated signatures had errors caught at these gates. This is not overhead; it is the difference between shipping broken code and shipping working code.

Lessons from Failure

We iterated hard. Here is what *did not* work initially.

- **Raw JSON to LLM:** Inconsistent schemas caused inconsistent outputs. $\rightarrow$ Built Behavior IR.
- **Fully automated ATT&CK mapping:** LLM hallucinated on subtle behavioral indicators. $\rightarrow$ Added human review with confidence thresholds.
- **No validation:** $\sim$ 20% of signatures had syntax or logic errors. $\rightarrow$ Implemented three-gate pipeline.

Key Lesson from Security Domain

In security automation, speed without verification is reckless. The slowdown from validation is not a limitation; it is a feature.

Solutions to Major Challenges

Challenge	Solution Implemented
Inconsistent CAPEv2 schemas	Generic Behavior IR parser with soft error handling
LLM output variation	Prompt templating, result caching, family-specific fine-tuning
Signature syntax errors	Automated Python syntax checker
Logic contradictions	Static behavior pattern analyzer
MITRE edge cases	Confidence thresholds + mandatory analyst review

Each challenge required incremental hardening, not a silver-bullet fix.

Core Performance Metrics

Precision	Recall	F1 Score
86%	79%	0.82

Quality Metrics

- False positive rate: 8%
- False negative rate: 21%
- Validation catches $\sim$ 15% of errors

Output Volume

- 20 signatures generated
- 10 malware families
- 35+ MITRE techniques

Generalization: Performance on Unseen Families

Does the system learn reusable behavior patterns?

Metric	Seen Families	Unseen Families
Precision	86%	77%
Recall	79%	65%

Interpretation

A 14% recall drop on unseen families is expected and acceptable. The system captures generalizable patterns, but analyst expertise remains essential for edge cases (particularly Discovery and Lateral Movement techniques).

MITRE ATT&CK Mapping Quality

Beyond signature generation, we evaluated mapping accuracy to the MITRE framework (84% overall).

Technique Category	Accuracy
Execution	91%
Persistence	88%
Command & Control	86%
Discovery	72%

Key Insight

Explicit behaviors (Execution, C2) map well. Implicit indicators (Discovery, lateral techniques) require human judgment. This asymmetry guides validation effort allocation.

Analyst Workload and Time Savings

- **Manual workflow (baseline):** 4–6 hours per sample
- **LLM draft:** 15–20 minutes (data extraction + generation)
- **Validation and refinement:** 40–50 minutes
- **Total with tool:** $\sim$ 1 hour per sample

Operational Impact

80%+ time reduction per sample. Analyst role shifts from signature author to reviewer and validator.

This is an analyst multiplier, not a replacement.

Five Key Takeaways

1. **LLM-driven signature synthesis is viable.** With proper abstraction and validation, we achieve deployment-grade precision.
2. **Behavior abstraction is non-negotiable.** The IR layer absorbs schema inconsistency and reduces hallucination. This investment pays dividends.
3. **Never trust LLM outputs unconditionally.** In security, three validation gates are not optional. Every signature needs syntax, logic, and human review.
4. **Generalization has real bounds.** Unseen families see a 14% recall drop. Diversity in training data matters for long-term use.
5. **The analyst doesn't disappear.** The system augments human judgment; it does not replace it. Role shifts from author to curator and validator.

What's Next

Immediate improvements (0–3 months)

- Expand dataset: 500+ samples, 30+ families for better generalization
- Reduce false negatives in Discovery/Lateral Movement via confidence-aware filtering
- Add automated regression tests to prevent deployed signature degradation

Medium-term roadmap (3–12 months)

- Fine-tune Llama on security-specific corpora
- Integrate with live SOC infrastructure (Splunk, ELK)
- Build feedback loop: production detections $\rightarrow$ signature refinement

Vision

A continuous learning system where analysts curate live detections, and the pipeline continuously improves signature quality in production.

Thank You

Questions?

Ikram Benfellah · Fadel Fatima Zahra

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